Oberwolfach Arbeitsgemeinschaft (ID: 2642)
New Homological Methods in Toric Geometry

Organizers: Michael Brown, David Eisenbud, and Daniel Erman

Sunday 11 October - Friday 16 October 2026

Schedule of talks:

no.	time	title of talk	name
1	Mo 9:00-9:50 am	Derived category of projective space, I	Amal Matoo
	Mo 10:00-10:30 am	Coffee break	
2	Mo 10:30-11:20 am	Derived category of projective space, II	Han Bom Moon
3	Mo 11:30-12:20 pm	Derived category of projective space, III	Isidora Bailly-Hall
4	Mo 3:30-4:20 pm	Toric varieties and Cox rings I	Sharon Robbins
	Mo 4:30-5:00 pm	Coffee break	
5	Mo 5:00-5:50 pm	Toric varieties and Cox rings II	Caitlin Davis
6	Tu 9:00-9:50 am	The Bondal-Thomsen Collection	Andreas Hochenegger
	Tu 10:00-10:30 am	Coffee break	
7	Tu 10:30-11:20 am	Constructing the Hanlon-Hicks-Lazarev resolution	Jason McCullough
	Tu 11:30-12:00 pm	Discussion: Computation	
8	Tu 3:30-4:20 pm	More examples and features of the HHL resolution	Ryan Watson
	Tu 4:30-5:00 pm	Coffee break	
9	Tu 5:00-5:50 pm	Proof of the HHL Theorem	Greg Smith
10	We 9:00-9:50 am	Applications of the HHL Theorem	Jay Yang
	We 10:00-10:30 am	Coffee break	
11	We 10:30-11:20 am	The secondary fan	Mahrud Sayrafi
12	We 11:30-12:20 pm	The Cox category, I: Definition	Martin Kalck
	We Afternoon	Free for hikes etc.	
	We 7:30-8:30 pm	Discussion: Toric stacks	
13	Th 9:00-9:50 am	The Cox category, II: Fourier-Mukai transforms	John Cobb
	Th 10:00-10:30 am	Coffee break	
14	Th 10:30-11:20 am	Proof of revised King's conjecture, I: Outline	Milena Hering
	Th 11:30-12:00 pm	Open Q&A	
15	Th 3:30-4:20 pm	Proof of revised King's conjecture, II: Exceptionality	Patrick Kennedy-Hunt
	Th 4:30-5:00 pm	Coffee break	
16	Th 5:00-5:50 pm	Proof of revised King's conjecture, III: Conclusion	John Nolan
17	Fri 9:00-9:50 am	Applications of Cox categories	Dashiell Stevanovich
	Fri 10:00-10:30 am	Coffee break	
18	Fr 10:30-11:20 am	Tate resolutions over projective space	Matthias Zach
19	Fr 11:30-12:20 pm	Tate resolutions over toric varieties	Sasha Zotine

Arrival day: Sunday afternoon/evening
Breakfast 8-9 am, lunch 12:30-1:30 pm, dinner 6:30-7:30 pm (In case of a public holiday such as Easter Monday, lunch/dinner take place in Hotel Hirschen and start at 1:15 pm / 6 pm respectively.)
Wednesday afternoon: traditional excursion
Departure day: Friday, before dinner-time